



Build your own Supercomputer for ₹0

...with the hardware you already have

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A supercomputer is something that is awe-inspiring; the stuff of dreams. We all want to see one, and sit and work on one. Why? What a silly question to ask a Digit reader. Now you could break into some defence lab and try playing chess against one – you'd lose the game and your right to live, but it might be worth it... you could also choose to save 10 years of your salary to blow up on 3 minutes with a supercomputer... or you could just whack your friends over the head with this book, quieten them down and read this article.

If you really want a supercomputing experience, all you need is a local area network, as many PCs as you can get your hands on, and this issue of Digit (yes the DVD too), and you're on your way.

So how super a computer can you throw together? If you're doing this at home, with say, two or three PCs, then you can set up something pretty decent,

but if you're in a college, or in a company with relaxed IT rules, you can easily touch Giga-flop territory; Tera-flops if it's a corporate office, and you'd probably be able to break world records if you happen to be network admin at Google or Microsoft. The beauty of it all is that we'll show you how you can use idle PCs, without changing any of the contents on the hard drives, or "ruining" anyone's work, or any such nonsense. All you need is a CD-drive on the server PC and network cards that are capable of booting from LAN, which basically any current generation LAN card / chip is capable of.

A short history lesson

Seymour Cray led the way in the 1960's and was at the forefront of design and development of supercomputers. We have come a long way since then, and supercomputers today perform calculation intensive tasks such as quantum physics, weather forecasting, wind tunnel research, simulation of detonation of nuclear bombs and many more such compute intensive tasks.

Why?

Why not? If the looks of awe you will get from friends aren't reason enough, consider these uses:

1.Render

If you're a 3ds Max junkie, a supercomputer would save you hours of boredom as you wait for renders to complete. You can use Autodesk Backburner in Windows to speed up your rendering over a cluster of home PCs. Those of you looking to do Maya rendering are also in luck, as there are render managers available – <http://bit.ly/dok3IZ>. If you are looking to do 3D content creation on your cluster, you have the option of using Blender (<http://www.blender.org>) which is a free and open source suite that is available for all major operating systems.

2.Donning a white hat

Brute-force hacking of passwords usually involves a large amount of computational power, or years of regular PC time. Not for you, right? Software